Souvik Samanta

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GeeksForGeeks | LinkedIn | GitHub | HackerRank | Portfolio

Experience

Tata Consultancy Services **System Engineer**
Nov 2024 – Present

Project: Agentic Incident-to-PR Remediation Platform *Google ADK, Vertex AI, Gemini, GCP, Python*

- Developed an **enterprise multi-agent GenAI system** automating the workflow from **production incident analysis to evidence-grounded code remediation and pull-request creation**.
- Designed a **Coordinator/Controller Agent** with specialized **Log and Gitea Agents** for incident investigation, evidence validation, repository navigation, and remediation.
- Integrated **Google Cloud Logging** and controlled Python tools for log querying, repository search, code modification, and PR creation, enforcing **least-privilege access**.
- Utilized **Gemini CLI** to analyze code and generate minimal, evidence-grounded patches, with mandatory **human-in-the-loop PR review**.
- Implemented **PII sanitization, prompt-injection protection, sandboxed execution, private connectivity, non-root containers, and vulnerability scanning**; achieved **Top 10 in the TCS Google Hackathon 2026** and presented the solution to **Google**.

Project: Financial Document Extraction and Intelligence *Azure Document Intelligence, Azure OpenAI, GenAI*

- Developed a **GenAI-powered financial document processing solution** automating extraction and intelligence from **Balance Sheets, Income Statements, and Cash Flow Statements**.
- Integrated **Azure Document Intelligence and Azure OpenAI/GenAI** to extract financial tables, contextual relationships, metadata, hierarchies, and mathematical relationships from unstructured PDFs.
- Developed support for **multi-page, multidimensional, image-based, and native-language documents**, including multilingual processing and financial metadata such as currency, units, periods, separators, and negative values.
- Designed **human-in-the-loop validation, automated quality checks, supplement-data processing, and CSV/ZIP output** to reduce manual validation and downstream processing effort.
- Performed **GenAI fine-tuning experiments**, achieving **100% Cash Flow extraction accuracy on a 10-record test set**; designed the solution using enterprise Azure services including **App Service, API Management, Functions, Key Vault, Entra ID, Private Link, Monitor, Log Analytics, Defender, and Content Safety**.

Celebal Technologies **Intern**
May 2023 – Jul 2023 **Internship Certificate**

- Selected from **20,000 applicants**; utilized **Azure Data Factory** to extract API data and load it into **Azure Data Lake Storage**.
- Architected **Apache Spark** transformations that improved data quality and reduced data retrieval time by **60%**.

Education

Techno Main Salt Lake Kolkata, India
Computer Science and Engineering, B.Tech Aug 2020 – Jul 2024
CGPA: 8.82

Technical Skills

Languages: Python, C++, Java, JavaScript **AI/GenAI:** LLMs, RAG, Multi-Agent Systems

Frameworks: LangChain, LangGraph, Google ADK

Cloud: GCP, Vertex AI, Azure, Azure OpenAI, Azure Document Intelligence

Tools: Git, GitHub, Docker, Kubernetes, Databricks, REST APIs

Certifications

- Programming, Data Structures and Algorithms Using Python by IIT Madras – **NPTEL – IIT Madras**
- Oracle Cloud Infrastructure 2023 Certified Foundations Associate – **Oracle**

Honors & Awards

- **Winner** of the **TCS AI Hackathon**.
- **3rd Place** in the **TCS CMI Unit CodeMaster Contest**.
- **Top 10** in the **TCS Google Hackathon 2026**; presented the solution to **Google**.
- **3rd Place** in the **ACM India East Bachelor Dissertation Award 2024**, competing against **200+ colleges**.
- Participated in the **Google Cloud Ready Facilitator Program 2022**; qualified for **Amazon ML Summer School 2023** among **40,000+ applicants**.